Topic 00: Infosys Puzzle Solving Master Formulas, Axioms Shortcuts Cheatsheet

> **Target Exam:** Infosys System Engineer (SE), Digital Specialist Engineer (DSE), Specialist Programmer (SP)
> **Domain:** Mathematical Puzzles, Matrix Grids, Geometric Figures, Alphabet Positional Reasoning
> **Status:** Production-Ready Engineering Cheatsheet

1. Executive Summary Core Axioms

In the Infosys recruitment examination, the **Puzzle Solving** section assesses a candidate's high-speed algebraic pattern recognition, discrete logic deduction, and spatial arithmetic acuity. Puzzles are presented in two distinct formats:
1. **Matrix Number Grids:** $3 × 3$, $4 × 3$, or $4 × 4$ numeric matrices with one or more unknown values (`?` , `X` , `Y`).
2. **Geometric Polygon Figures:** Triangles, circles, propellers, ovals, and multi-segmented boxes with peripheral inputs mapping to an internal focal value.

The Fundamental Axiom of Pattern Invariance

Every valid puzzle in the Infosys examination satisfies a **single, deterministic algebraic rule** across all reference components (rows, columns, or figures). If a candidate rule satisfies component 1 but fails component 2, it is categorically false.

2. Matrix Grid Solving Archetypes Mathematical Formulations

Archetype A: Linear Row Column Combinations

Pattern Name	Mathematical Formulation	Typical Manifestation
Row-wise Summation	$C_{i,}3 = C_{i,}1 + C_{i,}2$	The third column is the direct sum of the first two.
Row-wise Difference	$C_{i,}3 = C_{i,}1 - C_{i,}2$	The third column is the arithmetic difference.
Incremented Product Sum	$C_{i,}3 = (C_{i,}1 + 1)(C_{i,}2 + 1) - 1$	$C_3 = C_1 C_2 + C_1 + C_2$.
Linear Differential Balance	$R_1 - R_2 = R_4 - R_3$	Constant gap across paired alternating rows.
Linear Multi-Row Superposition	$R_4 = R_2 + R_3 - R_1$	Column accumulation with subtractive offset.

Archetype B: Multiplicative Fractional Proportions

Pattern Name	Mathematical Formulation	Typical Manifestation
Direct Product	$C_3 = C_1 \times C_2$	Direct multiplicative factor.
Product over Divisor	$C_4 = (C_1 \times C_2) / C_3$	Cross-column quotient balancing.
Proportional Scaling	$R_k = \lambda \cdot R_k\text{-}1$	Row 2 is $0.5 \times R_1$; Row 3 is $3 \times R_2$.
Column Accumulation Scaling	$C_4 = \lambda \sum[j=1 \text{ to } 3] C_j$	Fourth column is a scalar multiple of column sum (e.g. $\lambda = 2$).
Fractional Mapping	$C_3 = C_1 \times p / q$	Fractional scalar producing an integer mapped to the alphabet.

Archetype C: Powers, Exponents Polynomial Grids

Pattern Name	Mathematical Formulation	Typical Manifestation
Exponential Degrees	$M_{i,}j = a_i\text{^}j$	Consecutive powers: a^1, a^2, a^3, a^4.
Square Offset	$C_3 = C_2 + C_1^2$	Column 3 is Column 2 plus square of Column 1.
Sum of Squares	$C_3 = C_1^2 + C_2^2 - 1$	Sum of squares with fixed decremental offset.
Radical / Cube Root Sum	$\sqrt[3]{C_1} + \sqrt[3]{C_2} = C_3$	Sum of integer cube roots yields target integer.
Squared Differences	$C_3 = (C_1 - C_2)^2$	Square of difference between first two columns.

3. Geometric Figure Puzzle Formulations

1. Triangles with Vertex Inputs

- **Additive Factor:** $Center = (Top + Bottom-Right) \times Bottom-Left$
- **Digital Sum of Product:** $Center = DigitalRoot(V_1 \times V_2 \times V_3)$
- **Pure Product Factor:** $Center = V_1 \times V_2 \times V_3$

2. Segmented Polygons Ovals

- **Cross Determinant:** $Center = (Left \times Right) - (Top \times Bottom)$
- **Propeller / Cross Product:** $Center = (TL \times BR) + (TR \times BL)$
- **Corner Degree Superposition:** $Center = TR^3 + TL^2$
- **Semiproduct with Constant Offset:** $Center = (Left \times Right) / 2 - k$

4. Alphabet-to-Number Positional Logic ($A=1 \dots Z=26$)

In Infosys puzzles, alphabets represent their forward English positional values:
$$
A=1 B=2 C=3 D=4 E=5 F=6 G=7 H=8 I=9

J=10 K=11 L=12 M=13 N=14 O=15 P=16 Q=17 R=18

S=19 T=20 U=21 V=22 W=23 X=24 Y=25 Z=26
$$

Common Alphabet Transformations

1. **Positional Difference Doubling:** $(Top - Bottom) × 2 = Center$.
2. **Alphabet Arithmetic Progression:** Odd sequences ($1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23$).
3. **Paired Letter Multiplicative Sum:** $(L_1 × L_2) + (L_3 × L_4) = Number$.
4. **Alphabet Semi-Product:** $(L_1 × L_3) / 2 = L_2$.
5. **Sum Difference Vector:** $L_1 - L_2 = C_2$ and $L_1 + L_2 = C_3$.

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5. Strategic 60-Second Solving Algorithm for Test Takers

When facing an unknown matrix or figure puzzle under test conditions, apply this prioritized sequence:

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[Step 1: Check Magnitude Gradient]
├── Are numbers roughly similar?
│   └── Test Row/Column Sums, Differences, or Averages.
└── Are numbers exploding in size?
    └── Test Multiplication, Squares, Cubes, or Powers.

[Step 2: Check Dimensions Symmetry]
├── 3x3 Grid: Check Row operations first, then Column operations, then Diagonals.
├── 4x3 Grid: Check if Row 4 is the sum or multiple of Rows 1, 2, 3.
└── 4x4 Grid: Check for parallel Arithmetic Progressions along rows and columns.

[Step 3: Test Digits Non-Linear Functions]
├── Do numbers resist standard algebra?
│   ├── Test Digit Products: (Digit 1 * Digit 2) + ...
│   ├── Test Digital Root: Sum of digits until single digit.
│   └── Test Reverse Digits: |A - B| with reversed digit order.

[Step 4: Distractor Elimination]
├── Calculate parity (odd/even).
├── Check monotonic bounds.
└── Reject options that violate modular arithmetic.
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